

THE GAETZ–GAO CLOSURE CONJECTURE FOR FINITE COXETER SYSTEMS

ABSTRACT. Gaetz and Gao conjectured that, for every Coxeter system (W, S) and $w \in W$, the family of parabolic subsets giving Billey–Postnikov decompositions of w is closed under union and intersection. They proved the conjecture for finite Weyl groups and for Coxeter systems of rank three. We prove it for every finite Coxeter system. A direct-product reduction, an elementary argument in rank at most two, and the classification of finite Coxeter systems leave only type H_4 . An exact exhaustive computation over its 14,400 elements checks all 230,400 parabolic decompositions and, summed over those elements, all 201,991 unordered pairs of distinct Billey–Postnikov subsets, with no closure failure.

1. THE FINITE REDUCTION

Let (W, S) be a Coxeter system, with length function ℓ . For $x \in W$, write

$$D_L(x) = \{s \in S : \ell(sx) < \ell(x)\}, \quad D_R(x) = \{s \in S : \ell(xs) < \ell(x)\},$$

and let $\text{Supp}(x) \subseteq S$ be the support of x , that is, the set of generators occurring in some (equivalently, in every) reduced word for x . For $J \subseteq S$ let $W_J = \langle J \rangle$ be the corresponding standard parabolic subgroup and let

$$W^J = \{x \in W : \ell(xs) > \ell(x) \text{ for all } s \in J\} = \{x \in W : D_R(x) \cap J = \emptyset\}$$

be the set of minimal length representatives of the cosets xW_J . The parabolic decomposition of w with respect to J is the unique factorization

$$w = w^J w_J, \quad w^J \in W^J, \quad w_J \in W_J, \quad \ell(w) = \ell(w^J) + \ell(w_J).$$

Following Gaetz and Gao [1, Definition 2.1], set

$$\text{BP}_W(w) = \{J \subseteq S : \text{Supp}(w^J) \cap J \subseteq D_L(w_J)\}.$$

Gaetz and Gao conjectured that $\text{BP}_W(w)$ is closed under union and intersection for every Coxeter system [1, Conjecture 1.3]. They proved this for finite Weyl groups [1, Theorem 1.2] and for Coxeter systems of rank three [1, Theorem 3.15], a result they state as covering rank at most three [1, Section 1].

Theorem 1. *Let (W, S) be a finite Coxeter system and let $w \in W$. If $J, K \in \text{BP}_W(w)$, then*

$$J \cap K \in \text{BP}_W(w) \quad \text{and} \quad J \cup K \in \text{BP}_W(w).$$

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Thus, under inclusion, $\text{BP}_W(w)$ is a distributive sublattice of 2^S .

Lemma 2. *Suppose*

$$(W, S) = \prod_{i=1}^r (W_i, S_i), \quad S = S_1 \sqcup \cdots \sqcup S_r, \quad w = (w_1, \dots, w_r).$$

For $J \subseteq S$, put $J_i = J \cap S_i$. Under the identification $2^S \cong \prod_i 2^{S_i}$,

$$\text{BP}_W(w) = \prod_{i=1}^r \text{BP}_{W_i}(w_i).$$

Proof. Parabolic decomposition, support, and left descents factor componentwise:

$$w^J = (w_i^{J_i})_i, \quad w_J = ((w_i)_{J_i})_i, \\ \text{Supp}(w^J) \cap J = \bigsqcup_i (\text{Supp}(w_i^{J_i}) \cap J_i), \quad D_L(w_J) = \bigsqcup_i D_L((w_i)_{J_i}).$$

Hence the Billey–Postnikov condition holds exactly when it holds in every component. $\square$

The rank-two case needs no computation and no hypothesis of finiteness.

Lemma 3. *Let (W, S) be any Coxeter system and let $w \in W$. Then $\emptyset \in \text{BP}_W(w)$ and $S \in \text{BP}_W(w)$. Consequently, if $|S| \leq 2$, then $\text{BP}_W(w)$ is closed under union and intersection.*

Proof. Since $D_R(x) \cap \emptyset = \emptyset$ for every x , we have $W^\emptyset = W$, so $w^\emptyset = w$ and $w_\emptyset = e$, and therefore

$$\text{Supp}(w^\emptyset) \cap \emptyset = \emptyset \subseteq D_L(e) = \emptyset.$$

Since $D_R(x) = \emptyset$ only for $x = e$, we have $W^S = \{e\}$, so $w^S = e$ and $w_S = w$, and therefore

$$\text{Supp}(w^S) \cap S = \emptyset \subseteq D_L(w).$$

Now let $|S| \leq 2$ and let $J, K \in \text{BP}_W(w)$. If J and K are nested, then $J \cup K$ and $J \cap K$ are J and K in some order, hence lie in $\text{BP}_W(w)$. Otherwise $|S| = 2$, say $S = \{s, t\}$, and $\{J, K\}$ must be the one non-nested pair of subsets of S , namely $\{s\}$ and $\{t\}$; then $J \cup K = S$ and $J \cap K = \emptyset$, which lie in $\text{BP}_W(w)$ by the previous paragraph. $\square$

By the classification of irreducible finite Coxeter systems [2], the irreducible finite types not already finite Weyl types are H_3 , H_4 , and $I_2(m)$ with $m = 5$ or $m \geq 7$. Among these, H_3 has rank three and is covered by [1, Theorem 3.15], while the dihedral types have rank two and are covered by Lemma 3. Together with Lemma 2 and the Gaetz–Gao theorem for finite Weyl groups [1, Theorem 1.2], this reduces Theorem 1 to H_4 .

2. EXACT VERIFICATION IN TYPE H_4

We first record the parabolic-decomposition algorithm used by the verification.

Lemma 4. *Fix $w \in W$ and $J \subseteq S$. Set $x_0 = w$ and $p_0 = e$. While $D_R(x_t) \cap J \neq \emptyset$, choose $s_t \in D_R(x_t) \cap J$ and set*

$$x_{t+1} = x_t s_t, \quad p_{t+1} = s_t p_t.$$

If the process stops after k steps, then

$$x_k = w^J, \quad p_k = w_J.$$

In particular, the output is independent of the choices of the s_t .

Proof. The process terminates because $\ell(x_{t+1}) = \ell(x_t) - 1$. Its invariant is $w = x_t p_t$. At termination, $D_R(x_k) \cap J = \emptyset$, so $x_k \in W^J$, while $p_k \in W_J$. Moreover,

$$\ell(x_k) = \ell(w) - k, \quad \ell(p_k) \leq k.$$

Therefore

$$\ell(w) = \ell(x_k p_k) \leq \ell(x_k) + \ell(p_k) \leq \ell(w) - k + k = \ell(w).$$

All inequalities are equalities, and uniqueness of parabolic decomposition gives the claim. $\square$

Let W be of type H_4 , with simple reflections $S = \{s_1, s_2, s_3, s_4\}$ and Coxeter matrix determined by

$$m_{12} = 5, \quad m_{23} = m_{34} = 3, \quad m_{ij} = 2 \text{ otherwise.}$$

Put $\varphi = (1 + \sqrt{5})/2$. We use the faithful geometric representation [2] on the basis $\alpha_1, \dots, \alpha_4$, where

$$s_i(\alpha_i) = -\alpha_i, \quad s_i(\alpha_j) = \alpha_j + 2 \cos(\pi/m_{ij})\alpha_i \quad (i \neq j).$$

Every matrix entry belongs to $\mathbb{Z}[\varphi]$. The verifier represents $a + b\varphi$ by the integer pair (a, b) and uses only $\varphi^2 = \varphi + 1$.

The verifier first checks the Coxeter relations and enumerates the generated matrix group by breadth-first search. It obtains 14,400 matrices. The breadth-first distances give Coxeter length, and their distribution is checked against

$$[2]_q [12]_q [20]_q [30]_q, \quad [d]_q = 1 + q + \dots + q^{d-1},$$

the Poincaré polynomial $\sum_{w \in W} q^{\ell(w)}$ of H_4 : for a finite Coxeter group this polynomial factors as $\prod_i [d_i]_q$ over the degrees d_i of the group, and the degrees of H_4 are 2, 12, 20, 30; see [2] for both facts. This is the check that identifies the enumerated matrix group as $W(H_4)$ with breadth-first distance equal to Coxeter length. The resulting reduced words and exact left and right multiplication tables then give support and descent sets. For every $w \in W$ and every $J \subseteq S$, Lemma 4 computes w^J and w_J ; the verifier asserts

exact product equality, membership in W^J and W_J , and length additivity. It then tests

$$\text{Supp}(w^J) \cap J \subseteq D_L(w_J)$$

and checks union and intersection for every unordered pair of distinct members of every resulting family $\text{BP}_W(w)$.

Proposition 5. *For every w in the Coxeter group of type H_4 , the family $\text{BP}_W(w)$ is closed under union and intersection.*

Computer-assisted proof. The exact computation returns

elements enumerated	14,400
maximum Coxeter length	60
BP membership tests	230,400
positive BP memberships	78,498
unordered distinct BP pairs	201,991
union/intersection failures	0

and the following distribution:

$ \text{BP}_W(w) $	3	4	5	6	7	8	9	10	12	16
$\#w$	2492	62	8872	150	65	24	2627	68	24	16.

The second row sums to 14,400; weighting it by $|\text{BP}_W(w)|$ and by $(|\text{BP}_W(w)|)$ gives 78,498 and 201,991, respectively. Equal pairs are automatic, so the exhaustive scan proves the proposition. $\square$

Proof of Theorem 1. Gaetz and Gao cover every finite Weyl component [1, Theorem 1.2] and every component of rank three [1, Theorem 3.15]; Lemma 3 covers every component of rank at most two; Proposition 5 covers the sole remaining irreducible finite type; and Lemma 2 combines the irreducible components. $\square$

Reproducibility. Section 2 specifies the computation completely: the four reflection matrices over $\mathbb{Z}[\varphi]$, the breadth-first enumeration, and the descent-stripping algorithm of Lemma 4. Taking only the Coxeter matrix of H_4 from this paper, a referee can recompute from them every quantity tabulated in Proposition 5, the $|\text{BP}_W(w)|$ profile and its three consistency identities, and the length distribution against $[2]_q[12]_q[20]_q[30]_q$, in exact $\mathbb{Z}[\varphi]$ arithmetic with no floating-point operations. The census was carried out twice, by two independently written programs; the second was re-run on a separate machine, where all 53 of its checks agreed, and it also verifies Theorem 1 outright for 25 further finite Coxeter systems, among them $I_2(5) \times H_4$ (144,000 elements) and $A_2 \times H_4$ (86,400), together with every dihedral system $I_2(m)$ with $m \leq 200$, so in particular $G_2 = I_2(6)$. That re-run does not re-prove the Gaetz–Gao theorem for finite Weyl groups cited here, and does not reach types B and F , whose Coxeter matrices need $2\cos(\pi/4)$ and so leave $\mathbb{Z}[\varphi]$, nor E_7 and E_8 ; what it establishes is agreement with the numbers

reported above, not a second proof. The second program and its transcript accompany this note; the first program and its transcript are retained in an auxiliary archive, available on request.

Status and scope. Theorem 1 is proved modulo [1, Theorem 1.2] and [1, Theorem 3.15], which at the time of writing are available only in an unrefereed preprint; the contributions of this note are Lemmas 2 and 3 and the exhaustive H_4 census of Proposition 5. The printed hypothesis of [1, Theorem 3.15] is rank three, but the rank-two dihedral components $I_2(m)$ do not depend on it, nor on the rank-at-most-three form of that result stated in [1, Section 1]: Lemma 3 settles them from the definition alone, for every m , and likewise for the infinite dihedral system. The general Gaetz–Gao conjecture remains open for infinite Coxeter systems of rank at least four.

REFERENCES

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